# Lesson 8: Where Does the Wasted Energy Go?

## Unit

Energy: Stores and Transfers

## Lesson focus

Students learn that in every real change some energy is dissipated to the thermal store of the surroundings. They explain how lubrication and thermal insulation reduce unwanted transfers, test insulating materials with cooling cups, and apply the ideas to designing an energy-efficient house.

## Specification

AQA GCSE Physics 4.1.2.1 Energy transfers in a system. Students do not need the formal definition of thermal conductivity. The cooling cups practical is required practical 2 in GCSE Physics (8463) only; it is not in Combined Science: Trilogy (8464).

## Learning objectives

By the end of the lesson, students should be able to:

1. Explain what it means for energy to be dissipated, and where it ends up.
2. Explain how lubrication reduces unwanted energy transfers.
3. State that a material with a higher thermal conductivity transfers energy by conduction faster.
4. Explain how the thickness and thermal conductivity of a building's walls affect how fast it cools.
5. Suggest and justify ways of reducing unwanted energy transfers in a home.

## Key vocabulary

| Keyword | Student-friendly definition |
|---|---|
| Dissipated | Spread out into the surroundings, where it is less useful; often called "wasted" |
| Friction | A force that opposes motion when surfaces rub; it heats them |
| Lubrication | Putting oil or grease between moving parts to reduce friction |
| Conduction | Energy transferred through a material by heating, from the hotter part to the cooler part |
| Thermal conductivity | How well a material conducts; higher conductivity means faster conduction |
| Thermal insulator | A material with a low thermal conductivity |

## Equipment and resources

- Starter: two toy cars or bicycle wheels, one with oiled axles
- Per group: four identical cups or beakers, thermometers or temperature probes, stopwatch, measuring cylinder, lids
- Insulating materials: bubble wrap, cotton wool, newspaper, felt (one cup left bare as the control)
- Hot water from a kettle, cooled to about 60 °C and poured by the teacher
- Elastic bands to hold the wrappings
- Optional demonstration: rods of different metals with wax and drawing pins along them, heated at one end

## Safety notes

- The teacher pours the hot water. Use water at about 60 °C, not boiling.

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- Stand cups in a tray; keep the bench clear.
- Report spills and scalds at once; run cold water over any scald.

## Starter: Back to Lesson 6

Remind students that in Lesson 6 the pendulum never swung back to its release height. Ask: where did the energy go?

Rub hands together hard. Then spin two wheels, one with oiled axles. Which spins for longer, and why?

> *Friction transferred energy to the thermal store of the moving parts and the surroundings. The energy is not destroyed, but it is spread out and we cannot get it back to use. That is what "dissipated" means.*

## Main teaching: Reducing unwanted transfers

| Unwanted transfer | How to reduce it | Why it works |
|---|---|---|
| Friction between moving parts | Lubrication with oil or grease | The surfaces slide more easily, so less energy is transferred to the thermal store |
| Air resistance on a vehicle | Streamlined shape | Less force from the air, so less work against it |
| Conduction through walls and windows | Thermal insulation | Materials with low thermal conductivity slow the transfer |

## Main teaching: Conduction and buildings

Optional demonstration: wax-and-pin rods. The pins fall off the copper rod first, then aluminium, then steel, then glass. Higher thermal conductivity, faster conduction.

Then the two rules for a building:

1. The **lower** the thermal conductivity of the walls, the **slower** the house cools.
2. The **thicker** the walls, the **slower** the house cools.

Trapped air is a very poor conductor. Many insulators (wool, foam, bubble wrap, loft insulation, double glazing) work because they trap air.

## Practical: Cooling cups (Triple only)

Investigation question: which material is the best thermal insulator?

| Variable | |
|---|---|
| Independent | Insulating material (categoric) |
| Dependent | Temperature drop in 10 minutes |
| Control | Volume of water, starting temperature, type of cup, thickness of wrapping, lid |

Method:

1. Wrap each cup in one layer of a different material; leave one bare.
2. The teacher pours 100 $cm^3$ of water at about 60 °C into each cup. Fit the lids.
3. Record the starting temperature, then the temperature after 10 minutes.
4. Calculate the temperature drop.

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Sample results:

| Material | Start / °C | After 10 min / °C | Drop / °C |
|---|---|---|---|
| None (control) | 60 | 47 | 13 |
| Newspaper | 60 | 52 | 8 |
| Bubble wrap | 60 | 54 | 6 |
| Cotton wool | 60 | 55 | 5 |

Students present the results in a bar chart (the independent variable is categoric: Unit 1, Lesson 2).

Conclusion: cotton wool was the best insulator here, because it traps the most air and so has the lowest thermal conductivity.

## Closing task: Design an Eco House

Students annotate the house outline in the workbook with at least four ways to reduce energy transfer out of the house. For each, they name the feature, the unwanted transfer it reduces, and why it works, using "thermal conductivity" or "thickness".

| Feature | Why it works |
|---|---|
| Loft insulation | Thick layer of trapped air, low thermal conductivity |
| Cavity wall insulation | Foam fills the gap between brick walls; low thermal conductivity |
| Double glazing | Trapped air or gas between panes |
| Thicker walls | Energy takes longer to conduct through |
| Draught excluders, carpets | Reduce transfer through gaps and floors |

If time is short, set the Eco House as prep and keep the cooling cups.

## Check for understanding

1. Where does dissipated energy end up?
2. Why does oiling a bicycle chain make cycling easier?
3. Copper has a higher thermal conductivity than glass. Which transfers energy by conduction faster?
4. Two houses have walls of the same brick. One wall is twice as thick. Which house cools more slowly?

## Plenary

Students write one sentence that uses all of: dissipated, thermal store, surroundings, insulation.

## Possible prep

Complete the Eco House. Stretch: estimate which feature would save the most energy in your own home, and why.

## Using the workbook

The Quick check (slide 62) has its own box before the Exam corner. Students answer each question, then check it as you reveal the answer.

## Teacher notes

1. **"Energy is lost."** It is not lost; it is dissipated. Allow "wasted" as everyday shorthand, but make students say where it goes.

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2 **"Insulators keep the cold out."** Insulation slows the energy transfer from hot to cold. In winter that is from the house outwards.

3 **"Metal feels cold, so it is cold."** Metal at room temperature feels cold because it conducts energy away from your hand quickly.

4 Convection and radiation are not needed here; conduction and the idea of trapped air are enough.

5 The cooling-cup practical takes about 15 minutes with setup. Start it early and teach the building rules while the cups cool.

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